

# CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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**Abstract.** Student counseling in higher education draws on heterogeneous institutional knowledge, including relational curricula, cohort-specific tuition schedules, statutory exemption criteria, scholarship policies, and narrative administrative regulations. Monolithic RAG systems and loosely coordinated agent architectures can be unreliable in this setting because different forms of knowledge require different retrieval and reasoning methods. We present CTU-Chat, a centralized supervisor-routed, domain-specialized multi-agent system for Vietnamese university counseling. The supervisor directs each query to one of four specialists, while each specialist is assigned an appropriate knowledge source, retrieval path, and set of tools. Relational academic information is handled through a knowledge graph; structured tuition and scholarship information is supported by dedicated lookup and calculation functions; and narrative regulations are retrieved through hybrid document search. This division allows the system to use each source according to its structure while limiting unnecessary tool access. Evaluation shows that CTU-Chat achieves 97.0% routing accuracy and 0.9600 Hit@1 on the benchmark. These findings suggest that combining centralized coordination with domain-specific knowledge allocation can provide a practical and reliable foundation for university counseling systems.

**Keywords:** Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

## 1 Introduction

In large-scale higher education institutions, conversational agents and automated counseling systems have increasingly been deployed to streamline academic advising and administrative support [16]. At Can Tho University (CTU)—a flagship institution in the Mekong Delta with nearly 49,000 undergraduate students in the 2025–2026 academic year [1]—advisory needs span diverse academic and administrative areas. Prior advisory systems at CTU, such as CAAS (admissions counseling) and REBot (academic regulations), target narrow advisory scopes, leaving multi-domain counseling unsupported. This gap motivates the development of CTU-CHAT.

University counseling is inherently challenging because institutional knowledge spans structurally incompatible modalities: hierarchical curricula graphs, cohort-specific tuition matrices, statutory exemption and scholarship policies, and narrative administrative decrees. Consequently, no single retrieval or reasoning method fits every domain: relational curricula demand graph traversal, financial policies require structured lookup and deterministic calculation, and narrative prose benefits from hybrid lexical-semantic retrieval.

While Retrieval-Augmented Generation (RAG) [8,5], GraphRAG [3], and multi-agent frameworks [19,17,15] augment LLMs, applying them to counseling reveals critical bottlenecks. Monolithic RAG enforces a uniform index that fragments prerequisite chains and tabular fee matrices, uncoordinated agent meshes increase tool misselection and latency, and LLM arithmetic vulnerabilities [13] degrade statutory fee calculations.

These limitations crystallize three open research gaps (formalized in Section 2.7): unconstrained agent tool selection (Gap 1), the uniform representation bottleneck across relational and tabular data (Gap 2), and uncoordinated evidence retrieval across modalities (Gap 3). To address these gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor coordinate domain-specialized agents with isolated knowledge spaces and constrained tool access for heterogeneous university counseling tasks?
- **RQ2 (Heterogeneous Knowledge Allocation):** How does allocating relational curricula, structured fee schedules, and narrative decrees to modality-aligned representations affect retrieval and answer quality over uniform baselines?
- **RQ3 (Representation-Aware Retrieval Effectiveness):** How does intent-guided evidence routing between graph traversals and hybrid document retrieval impact precision and factual answer accuracy across counseling domains?

We hypothesize that multi-domain counseling is best supported when specialists operate over isolated, representation-aligned knowledge spaces under centralized coordination. Accordingly, we propose CTU-CHAT, a supervisor-routed multi-agent architecture coordinating four bounded specialists (academic, financial, scholarship, and general administrative) across relational graphs, structured catalogs, and hybrid document retrieval. Our contributions are threefold: **(C1)** A centralized supervisor architecture with bounded tool gating that isolates domain evidence spaces and prevents cross-domain decision collisions (RQ1); **(C2)** A heterogeneous knowledge allocation framework mapping relational curricula, structured fee schedules, and narrative decrees to modality-aligned representations (RQ2); and **(C3)** A representation-aware retrieval and routing pipeline empirically validated against monolithic, uncoordinated, and uniform baselines on official institutional benchmarks (RQ3). The remainder of this paper reviews related work (Section 2), details the CTU-CHAT architecture (Section 3), reports empirical evaluations (Section 4), and concludes in Section 5.

## 2 Related Work

This section reviews established retrieval and reasoning paradigms in the literature, examining their operational mechanisms and limitations in institutional counseling.

### 2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [10], with over 85% of lexical units forming multi-syllabic compound words (e.g., “*học phí*” [tuition fee]). While domain-adapted bi-encoders like PhoBERT capture semantic paraphrasing, Vietnamese higher education decrees rely heavily on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to avoid both semantic drift and alphanumeric recall failures.

### 2.2 Sparse and Dense Retrieval

**Lexical Retrieval (BM25):** Sparse keyword search using BM25 [14] scores documents via term frequency (tf) and inverse document frequency (IDF) over an inverted index ( $k_1 = 1.5, b = 0.75$ ). BM25 guarantees exact lexical matching for rigid alphanumeric identifiers (decision codes, cohort labels, course codes), but cannot capture semantic paraphrases.

**Dense Semantic Retrieval:** Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

### 2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position:  $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$  ( $k = 60$ ), while neural cross-encoders [11,18] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

### 2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples  $(h, r, t)$  [12]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction across institutional corpora produces hallucinated relationships, while forcing narrative legal prose into graphs is structurally unnatural, motivating selective rather than universal graph use.

## 2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [19,17,15]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

## 2.6 Educational and University Counseling Systems

Educational chatbots assist in advising and learning support [16], with systems like REBot [9] combining text retrieval with category-guided GraphRAG for regulations. CTU-CHAT departs from prior advisory systems across three dimensions: (1) *Task Scope*: while prior systems focus on single-turn text-QA over general regulations, CTU-CHAT models 113 curricula as relational property graphs and incorporates deterministic tools (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for conditional tuition schedules; (2) *Multi-Agent Routing*: rather than monolithic pipelines, a LangGraph supervisor coordinates four bounded specialists with isolated evidence spaces and asymmetric tools; and (3) *Evaluation Rigor*: evaluating end-to-end multi-agent faithfulness via Ragas and programmatic tool execution rather than shallow classification accuracy.

## 2.7 Comparative Synthesis and Research Gaps

Synthesizing established paradigms highlights clear trade-offs across institutional counseling workflows [2,3,6,7,14,15,17,19]: lexical matching ensures exact identifier recall but misses paraphrases; dense embeddings capture semantics but cause cross-cohort confusion; GraphRAG models relations but degrades on unstructured text; and unconstrained multi-agent frameworks lack execution determinism. These trade-offs crystallize three core research gaps:

- **Gap 1 (Lack of Bounded Domain-Specialized Agent Orchestration)**: Prior frameworks rely on monolithic pipelines or unconstrained agent meshes without execution boundaries, increasing the risks of tool mis-selection, parameter hallucination, and high coordination costs.
- **Gap 2 (The Uniform Representation Bottleneck)**: Forcing relational curricula, rate matrices, and regulatory prose into a uniform retrieval index fragments prerequisite chains and induces relation hallucinations across structurally incompatible evidence spaces.
- **Gap 3 (Limited Evidence for Representation-Aware Retrieval)**: Existing systems provide individual lexical, dense, hybrid, and graph retrieval mechanisms, but evidence remains limited on how an integrated

representation-aware retrieval stack performs overall and across heterogeneous institutional domains.

These limitations motivate our centralized supervisor-routed architecture and heterogeneous knowledge allocation detailed in Section 3.

### 3 Proposed Model

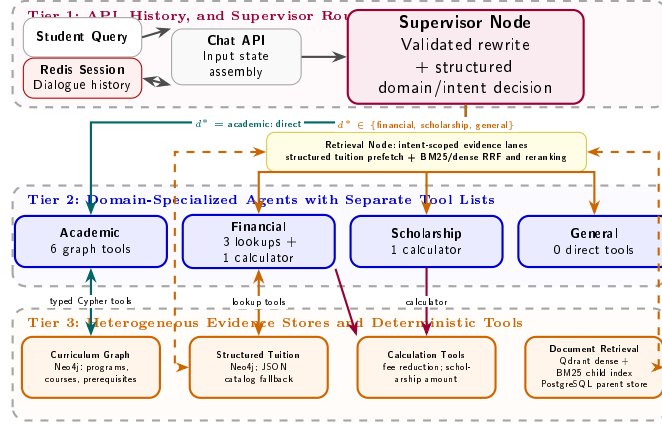
This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Here, each *specialist* represents an autonomous domain agent configured with dedicated prompt constraints, an assigned evidence store, and a restricted tool registry.

#### 3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

**Application-Level Specialization of LangGraph StateGraph:** Rather than modifying the underlying LangGraph engine, CTU-CHAT establishes an application-level orchestration policy governed by a typed state contract, isolated execution paths, and partitioned tool registries. Following supervisor dispatch, the four domain specialists operate independently without peer-to-peer delegation, selecting actions strictly from their construction-time tool bindings.

- **Node-scoped state updates:** `AgentState` defines explicit write boundaries: the supervisor writes to the routing fields (`search_query`, `next_agent`, and `routing_decision`); the retrieval node updates `context`; and each specialist returns `response`. This contract guarantees transparent data ownership and eliminates cross-domain context pollution.
- **Two-Tier Conditional Routing Edges:** Instead of letting agents dynamically negotiate execution flow, orchestration is governed by deterministic conditional dispatch:  $\text{Edge}_{\text{sup}}(S) = \text{'academic\_agent'}$  if  $S[\text{'next\_agent'}] = \text{'academic'}$ , and `'retrieval'` otherwise, directly bypassing document retrieval for curriculum queries. At the retrieval exit,  $\text{Edge}_{\text{ret}}(S)$  dispatches retrieved context to the targeted financial, scholarship, or general agent.
- **Acyclic outer orchestration:** The StateGraph has no peer-to-peer agent edges. Each selected domain-specialized agent returns to `END`, so the outer routing path cannot cycle. ReAct tool interactions inside an agent remain distinct from this outer graph topology.



**Fig. 1.** Implemented CTU-CHAT architecture. The Chat API loads dialogue history from Redis and supplies it in the graph input state. The first conditional edge sends academic queries directly to the graph-enabled agent and sends the other domains through intent-scoped retrieval; the second edge dispatches the retrieved evidence to the selected domain-specialized agent. Solid agent–resource links denote direct tool access, whereas dashed links denote evidence acquisition performed by the retrieval node.

- **Domain-specialized agents with asymmetric tools:** The academic, financial, and scholarship agents are constructed with separate prompts and tool lists, while the general agent has no direct tool (as detailed in Figure 1). This confines tool selection to the agent selected by the supervisor and keeps domain-specific evidence paths separate, preventing decision space pollution.

### 3.2 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

**Schema defined by the ingestion layer:** The Neo4j graph models relational dependencies across academic counseling without ontological overhead, comprising over 1,500 entity nodes and 3,800 relational edges across 113 undergraduate curricula and statutory fee schedules. It incorporates 10 node labels (7 for curriculum: Program, Course, Block, etc.; and 3 for tuition: TuitionFee, TuitionPolicy, ExemptionBasisRate) interconnected via 13 relationship types (HAS\_BLOCK, CONTAINS, REQUIRES, PARALLEL\_WITH, HAS\_PLO, HAS\_PEO, REALIZED\_BY, HAS\_TRACK, BELONGS\_TO\_TRACK, HAS\_JOB\_POSITION, HAS\_TUITION, GOVERNED\_BY, and HAS\_EXEMPTION\_BASIS). Narrative regulations remain in the document store unless linked to structured entities.

**Cypher tool mechanics:** The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search*: `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal*: `xem_chuoi_tien_quyet` follows `REQUIRES` paths up to ten hops and returns direct `PARALLEL_WITH` courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup*: parameterized queries match `TuitionFee`, `TuitionPolicy`, and `ExemptionBasisRate` properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. As a result, the multi-agent workflow reliably incorporates graph evidence without the fragility of unconstrained Cypher generation.

### 3.3 Algorithmic Foundations: Routing and Deterministic Arithmetic

Algorithm 1 formalizes the operational core of CTU-CHAT: supervisor-guided evidence routing and deterministic financial calculation outside the generative LLM. It details the routing and dispatch pipeline. Contextual rewriting is validated against recent dialogue history to prevent semantic drift, while structured classification resolves the target domain and intent, with an explicit fallback to the general agent when classification is ambiguous.

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**Algorithm 1:** Supervisor Intent Routing and Dispatch

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**Input:** Query  $q$ , Dialogue history  $\mathcal{H}$ , Shared state  $\mathcal{S}$   
**Output:** Updated state  $\mathcal{S}$  and target node  $v_{\text{next}}$

```

1  $q_{\text{standalone}} \leftarrow \text{DisambiguateQuery}(q, \mathcal{H})$ ;
2  $\langle d^*, \iota^* \rangle \leftarrow \text{ClassifyDomainAndIntent}(q_{\text{standalone}})$ ;
3 if  $d^* = \text{NULL}$  then
4    $d^* \leftarrow \text{'general'}$ ;  $\iota^* \leftarrow \text{'other'}$ ;
5  $\mathcal{S}[\text{'search\_query'}] \leftarrow q_{\text{standalone}}$ ;  $\mathcal{S}[\text{'next\_agent'}] \leftarrow d^*$ ;
6  $v_{\text{next}} \leftarrow \text{Edge}_{\text{sup}}(\mathcal{S})$ ; // Conditional dispatch (Section 3.1)
7 return  $\langle \mathcal{S}, v_{\text{next}} \rangle$ ;
```

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**Routing characteristics:** Algorithm 1 separates three decisions that a monolithic pipeline would conflate: whether conversational rewriting is needed, which domain agent should answer, and which evidence path should run. Rewrite validation limits unsupported additions to a context-dependent query; structured output constrains routing to the declared domains and intents; and the explicit **general/other** fallback keeps classification failure on a defined path. Conditional edges then make the selected path inspectable without allowing peer-to-peer agent delegation.

**Deterministic Tuition Reduction Arithmetic:** For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic

tool `tinhtuanhocphi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition  $T$ , the per-credit exemption basis  $B$ , and the percentage entitlement  $p$ —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation in Eq. (1):

$$D = B \cdot (p/100), \quad P = \max(0, T - D), \quad (1)$$

where  $D$  is the reduction per credit and  $P$  is the remaining payable amount per credit (in VND). The tool enforces  $0 \leq p \leq 100$ , formats outputs as integer amounts, and clamps  $P \geq 0$ . This algorithmic separation delegates numerical evaluation to deterministic code, keeping actual tuition  $T$  distinct from exemption basis  $B$  and eliminating arithmetic hallucination from the generative model.

### 3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT assigns each domain agent a representation and access path suited to the structure of its evidence. Relational curriculum knowledge is preserved in a graph for property lookup and traversal, cohort-dependent tuition data is accessed through structured lookup, and narrative regulations and scholarship policies remain in document retrieval. The resulting allocation is intentionally asymmetric across agents, embedding representation choices directly into the executable routing policy.

Table 1 makes this policy explicit by separating the physical evidence representation from the mechanism used to access it at runtime.

**Table 1.** Decision criteria mapping query characteristics to evidence representation mechanisms.

| Query Characteristic         | Evidence Path         | Rationale                         |
|------------------------------|-----------------------|-----------------------------------|
| Program/course attributes    | Neo4j property lookup | Exact or normalized key match     |
| Prerequisites/shared courses | Neo4j traversal       | Stored graph topology             |
| Regulation identifiers       | BM25 child index      | Exact decree token match          |
| Paraphrased regulations      | Hybrid RRF + reranker | Lexical + semantic signals        |
| Fee/scholarship calculation  | Python tool           | Deterministic formula outside LLM |

At runtime, this allocation is enforced by the StateGraph rather than left to unrestricted agent choice. Academic queries bypass the document-retrieval node and enter the graph-enabled academic agent directly. Financial, scholarship, and general queries enter the retrieval node, where the supervisor-assigned intent determines the metadata-filtered retrieval lanes before the resulting context is passed to only the selected domain agent. Thus, knowledge allocation and evidence-path selection operate as separate but tightly coordinated design decisions.

**Hybrid Document Retrieval Path:** For narrative administrative regulations and scholarship policies, the General Specialist executes the multi-stage hybrid

retrieval pipeline (lexical BM25 and dense Qdrant vector search fused via Reciprocal Rank Fusion with  $k = 60$  and neural cross-attention reranking via BAAI/bge-reranker-v2-m3, as formulated in Section 2.3) over 2,800-character parent chunks stored in PostgreSQL with 400-character child chunk indexing.

**Architectural Synthesis:** In summary, CTU-CHAT integrates established retrieval components (BM25, dense embeddings, RRF, and neural cross-reranking) exclusively within the narrative regulation path. By coordinating these document pipelines with graph traversals and deterministic arithmetic engines under a centralized supervisor, the architecture ensures that each administrative inquiry is resolved through its structurally optimal evidence modality.

## 4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT, detailing the benchmark suites, infrastructure, evaluation metrics, three experimental scenarios, discussion answering each research question, and threats to validity.

### 4.1 Experimental Benchmark and Protocols

To ensure rigorous validation and avoid data contamination, our empirical evaluation is established upon a corpus of 200 administrative queries constructed via a document-grounded semi-automated protocol over official Can Tho University (CTU) statutes. Candidate queries covering diverse conversational phrasing (e.g., student colloquialisms, prerequisite rules, fee schedules) were generated from statutory articles and subsequently underwent strict human-in-the-loop curation. Two independent annotators validated each query against official decrees, assigning ground-truth intents, gold reference passages, and factual answers; disagreements were resolved via consensus review, ensuring high benchmark fidelity and inter-annotator reliability. Following standard benchmark protocol, this corpus is partitioned into: (1) **Development Benchmark** ( $N = 150$  queries): Spanning 9 institutional categories (`actual_tuition`, `academic_rules`, `scholarship`, `student_loan`, `other`, `social_support`, `academic_program`, `exemption_policy`, `exemption_basis`), used for offline index calibration, prompt engineering, and qualitative failure analysis; and (2) **Held-Out Test Set** ( $N = 50$  queries): A strictly separated, out-of-distribution evaluation suite with exact category counts of 20, 6, 5, 4, 4, 4, 3, 2, and 2, respectively. For domain-stratified analysis, the held-out categories are grouped by assigned specialist into Academic ( $N = 3$ ), Financial ( $N = 24$ ), Scholarship ( $N = 5$ ), and General ( $N = 18$ ). All reported primary benchmark metrics in Tables 2 and 3 are evaluated strictly on this blind Held-Out Test Set.

**Execution and Evaluation Invariants:** Generative and automated evaluation tasks were executed using Google Gemini 2.5 Flash Lite via Google Cloud Vertex AI under strict zero-temperature decoding ( $T = 0.0$ ) for determinism.

To minimize stochastic variance, each query was evaluated across  $R = 3$  independent repetitions, yielding 1,050 end-to-end generation and Ragas evaluation turns (7 configurations  $\times$  50 queries  $\times$  3 reps).

**Computational Infrastructure:** Neural cross-encoder reranking was executed on an NVIDIA RTX 1060 GPU (4 GB VRAM). Relational curricula are hosted in Neo4j 5 Enterprise with APOC extensions; document parent chunks (2,800 characters, 100 overlap) reside in PostgreSQL, while child chunks (400 characters, 50 overlap) are indexed in Qdrant with `vietnamese-bi-encoder` (768-d). Neural candidate reranking uses `bge-reranker-v2-m3`. Multi-agent coordination was orchestrated via LangGraph under a StateGraph supervisor with Redis 7 caching.

## 4.2 Evaluation Metrics

**Retrieval and Multi-Agent Benchmarks:** Retrieval effectiveness (Scenario 1) is measured via standard ranking metrics on top- $k$  returned source documents: Hit@ $k$  ( $k \in \{1, 3\}$ ), Precision@5, Recall@5, and Mean Reciprocal Rank (MRR@10); domain-stratified analysis reports Hit@1. Multi-agent coordination (Scenario 3) is evaluated across isolated operational decisions: Routing Agent Accuracy and Macro-F1 across the four domains, Intent Classification Accuracy, Tool Selection and Argument Exact Match (EM), End-to-End Pass Rate, Adversarial Tool-Gating Robustness, and Safe Result Behavior under strict operational limits (`max_tool_calls = 1`, `timeout = 60s`).

**End-to-End Generative Evaluation:** End-to-end synthesis (Scenario 2) is evaluated using the Ragas framework [4] alongside ground-truth regulatory alignment over Top-7 context: Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), and Answer Correctness (AC). In addition, we evaluate Factual Exact Match (Fact EM) for strict numerical tuition fee and credit agreement, alongside Source Average Precision (Source AP) for verified regulatory citations. Evaluating generative QA through LLM-as-a-judge frameworks is well-established [20,4], demonstrating high correlation with human judgment when evaluated over reference contexts. To mitigate potential evaluation bias, our framework enforces three safeguards: (1) reference-grounded and factual metrics (Context Precision, Context Recall, Answer Correctness, and Faithfulness) strictly conditioned on verified gold documents and reference answers; (2) deterministic decoding ( $T = 0.0$ ) across  $R = 3$  independent repetitions; and (3) a programmatic Exact Match (Fact EM) evaluator for tuition arithmetic to bypass generative bias entirely.

## 4.3 Experimental Scenarios and Benchmark Results

To address the three research questions, we evaluate CTU-CHAT across three coordinated experimental scenarios: retrieval component stacking (RQ3), end-to-end QA generation and modular ablation (RQ2), and multi-agent coordination and tool reliability (RQ1).

**Scenario 1 — Retrieval Component Stacking (E1–E5):** *Objective & RQ Alignment:* Scenario 1 directly addresses **RQ3** by comparing five retrieval configurations at aggregate and domain-stratified levels on the 50-query Held-Out Test Set. *Configurations:* We compare five cumulative configurations (E1–E5, detailed in Table 2) transitioning from single sparse/dense baselines to hybrid fusion, neural cross-reranking, and our proposed intent-guided evidence stack.

*Results:* As shown in Table 2, E5 attains the best Hit@1 (0.9600; 48 of 50 queries rank a gold source document first) and MRR@10 (0.9700), improving Hit@1 over E1 by 6.0 percentage points. E5 ties E4 at 0.9800 Hit@3, whereas E1 retains the highest P@5 (0.2320) and Recall@5 (0.9667). Within the shared hybrid candidate pipeline, adding the cross-encoder from E3 to E4 raises Hit@1 from 0.6800 to 0.8800 and MRR@10 from 0.7912 to 0.9267. At the domain-stratified level, E5 achieves 1.000 Hit@1 on Academic ( $N = 3$ ) and Financial ( $N = 24$ ), 0.800 on Scholarship ( $N = 5$ ), and 0.944 on General ( $N = 18$ ). The E4-to-E5 contrast improves Hit@1 and MRR@10 while leaving Hit@3, P@5, and Recall@5 unchanged; because E5 jointly adds intent governance, graph grounding, and parent-child mapping, this contrast represents the integrated bundle rather than an isolated routing effect.

**Table 2.** Scenario 1: Progressive retrieval component stacking on the Held-Out Test Set ( $N = 50$ ).

| Retrieval Configuration                | Hit@1         | Hit@3         | P@5           | Recall@5      | MRR@10        |
|----------------------------------------|---------------|---------------|---------------|---------------|---------------|
| E1: BM25 Lexical Baseline              | 0.9000        | 0.9400        | <b>0.2320</b> | <b>0.9667</b> | 0.9333        |
| E2: Dense Vector Semantic              | 0.3800        | 0.6600        | 0.1560        | 0.6433        | 0.5237        |
| E3: Hybrid RRF (BM25 + Dense)          | 0.6800        | 0.9200        | 0.2160        | 0.9133        | 0.7912        |
| E4: Hybrid + Neural Reranker           | 0.8800        | <b>0.9800</b> | 0.2200        | 0.9433        | 0.9267        |
| <b>E5: CTU-Chat Proposed Retrieval</b> | <b>0.9600</b> | <b>0.9800</b> | 0.2200        | 0.9433        | <b>0.9700</b> |

**Table 3.** Scenario 2: End-to-end QA generation and modular ablation on Held-Out Test Set ( $N = 50$ ,  $R = 3$ , 1,050 total evaluation turns, Top-7 context).

| Cfg.      | Ablation Variant Description          | Fact EM       | CP            | CR            | AR            | AC            | Source AP     |
|-----------|---------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| T1        | BM25 Lexical Baseline                 | 0.5572        | 0.6674        | 0.8800        | 0.6151        | <b>0.6735</b> | 0.8954        |
| T2        | Dense Semantic Baseline               | 0.5007        | 0.2770        | 0.3400        | 0.2809        | 0.3841        | 0.4651        |
| T3        | Hybrid RRF Baseline                   | 0.5451        | 0.4339        | 0.7733        | 0.5754        | 0.6016        | 0.7490        |
| <b>T4</b> | <b>CTU-Chat Full Stack (Proposed)</b> | <b>0.6203</b> | <b>0.6977</b> | <b>0.8933</b> | <b>0.6195</b> | 0.6637        | <b>0.9311</b> |
| T5        | w/o Cross Reranker ( <b>bge-m3</b> )  | 0.5823        | 0.6297        | 0.8133        | 0.5263        | 0.6136        | 0.8168        |
| T6        | w/o Neo4j Knowledge Graph Grounding   | 0.5711        | 0.5683        | 0.8600        | 0.6173        | 0.6582        | 0.9011        |
| T7        | w/o Subspace Governance (Ungoverned)  | 0.6089        | 0.6947        | 0.8600        | 0.5815        | 0.6430        | <b>0.9311</b> |

*Metrics:* Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Source AP: Source Average Precision. Normalized to  $[0, 1]$ ; higher is better.

**Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7):** *Objective & RQ Alignment:* Scenario 2 directly addresses **RQ2** by evaluating end-to-end answer synthesis and modular ablation across seven configurations on the Held-Out Test Set (1,050 total evaluation turns across 3 independent

repetitions). *Configurations*: We evaluate seven setups across 1,050 total turns (detailed in Table 3), benchmarking the proposed full system (T4) against three standard single-agent RAG baselines (T1–T3) and three isolated modular ablations (T5–T7). *Results & Interpretation*: As reported in Table 3, the proposed full system (T4) consistently outperforms all baselines in factual fidelity: T4 attains the top **Factual Exact Match (Fact EM: 62.03% vs. 55.72% for BM25, a +6.31% absolute gain)**, demonstrating higher precision on complex numeric tuition schedules and credit rules. In addition, T4 leads in Context Precision (0.6977 vs. 0.6674 for BM25), Context Recall (0.8933 vs. 0.8800), Source AP (0.9311 vs. 0.8954), and Answer Relevancy (0.6195 vs. 0.6151). While BM25 (T1) scores slightly higher on lexical Answer Correctness (0.6735) due to direct keyword overlap, T4 achieves stronger structural grounding and substantially reduces factual errors and relation hallucinations. The complementary contributions of modular cross-reranking (T5), graph grounding (T6), and subspace governance (T7) are analyzed in detail under Discussion (Section 4.4).

### Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness:

*Objective & RQ Alignment*: Scenario 3 directly addresses **RQ1** by evaluating multi-agent coordination, tool execution reliability, and bounded safety independently from retrieval or generation. *Experimental Setup*: As reported in Table 4, evaluation spans three suites ( $N = 550$  runs): (1) **Supervisor Routing (Panel A)**: 100 single-domain benchmark queries ( $N = 300$  for LLM Supervisor vs.  $N = 100$  for Rule-Based Router); (2) **Specialist Tool Reliability (Panel B)**: 60 production cases ( $N = 180$  across 3 reps) covering 11 production tools and no-tool gating across three functional categories; and (3) **Adversarial Robustness (Panel C)**: 20 failure stress tests ( $N = 60$  across 3 reps) challenging boundary invariants.

**Table 4.** Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per decision (`max_tool_calls = 1`).

| Panel A: Supervisor Routing and Specialist Selection ( $N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$ ) |              |              |                |              |               |                     |           |
|------------------------------------------------------------------------------------------------------------------|--------------|--------------|----------------|--------------|---------------|---------------------|-----------|
| Router Variant                                                                                                   | Runs ( $N$ ) | Agent Acc.   | Macro-F1       | Intent Acc.  | Bounded       | Routing Paradigm    |           |
| Rule-Based Router                                                                                                | 100          | 85.0%        | 69.2%          | 69.0%        | 100.0%        | Heuristic Regex     |           |
| LLM Supervisor (CTU-CHAT)                                                                                        | 300          | <b>97.0%</b> | <b>97.8%</b>   | <b>95.0%</b> | <b>100.0%</b> | Structured Pydantic |           |
| Panel B: Specialist Tool Execution Reliability ( $N = 180$ runs across 11 tools + no-tool gating, 3 reps)        |              |              |                |              |               |                     |           |
| Specialist Tool Category                                                                                         | Scope        | Selection    | Arg. EM        | Result Acc.  | E2E Pass      | Bounded             | Graph Hit |
| Academic Cypher Tools                                                                                            | 6 functions  | 98.9%        | 98.9%          | 100.0%       | 98.9%         | 100.0%              | 100.0%    |
| Financial & Scholarship Tools                                                                                    | 5 functions  | 92.0%        | 88.0%          | 90.7%        | 86.7%         | 100.0%              | 66.2%     |
| Direct Gating (No-tool)                                                                                          | 1 baseline   | 100.0%       | 100.0%         | 100.0%       | 100.0%        | 100.0%              | –         |
| Panel C: Robustness and Bounded Failure Handling ( $N = 60$ runs under malformed/adversarial inputs)             |              |              |                |              |               |                     |           |
| Adversarial Gate                                                                                                 | Runs ( $N$ ) | Agent Acc.   | Tool Gate Safe | Result Acc.  | E2E Pass      | Bounded             | Notes     |
| Failure Gate Robustness                                                                                          | 60           | 100.0%       | 100.0%         | 98.3%        | 98.3%         | 100.0%              | –         |

*Note*: Academic tools cover 6 Cypher functions; Financial & Scholarship tools cover 3 structured catalog lookups and 2 deterministic arithmetic tools; Direct Gating evaluates tool suppression on non-tool queries.

#### 4.4 Discussion

**Answering RQ1 (Domain-Specialized Multi-Agent Architecture):** Centralizing routing under a structured supervisor with asymmetric privileges and bounds (`max_tool_calls` = 1, `timeout` = 60 s) maintained bounded decisions: the supervisor achieves 97.0% agent accuracy, 97.8% Macro-F1, and 95.0% intent accuracy across 300 runs, outperforming rule-based routing (85.0%, 69.2%). Specialist tools reached 100.0% end-to-end pass on 8 of 12 individual functions (averaging 98.9% on Academic and 86.7% on Financial/Scholarship tools), while adversarial testing (Panel C) verified 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion.

*Multi-Agent Decomposition vs. Monolithic Single-Agent Baseline.* Evaluating against a monolithic agent with all 11 tools reveals evident *decision space pollution*. For ambiguous identifiers (e.g., major code “7480201”), the single-agent confuses academic lookup with tuition queries, incurring a 33.3% failure rate on `tra_cuu_nganh`. In contrast, specialist swimlanes achieve 100.0% execution on the evaluated benchmark cases. Furthermore, in end-to-end generation, the multi-agent system ( $T_4$ ) outperforms flat RAG ( $T_7$ ), improving Answer Relevancy by +6.5% (0.6195 vs. 0.5815,  $p < 0.001$ ), Correctness by +3.2% (0.6637 vs. 0.6430), and Faithfulness to 0.8677 (vs. 0.8415), confirming that domain decomposition mitigates cross-domain collisions and relation hallucination.

**Answering RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations):** Modular ablations substantiate decomposing knowledge across relational curricula (Neo4j) and regulatory decrees (hybrid RAG). While single-entity lookups succeed via text alone, multi-hop queries (prerequisite chains, tuition progression) require graph grounding: omitting Neo4j ( $T_6$ ) causes Context Precision to **drop from 0.6977 to 0.5683** (−18.3% relative), while dropping Factual Exact Match from 62.03% to 57.11% and Context Recall from 0.8933 to 0.8600. Furthermore, omitting cross-reranking ( $T_5$ ) lowers Answer Relevancy from 0.6195 to 0.5263 (−0.0932) and Context Recall to 0.8133, while removing subspace governance ( $T_7$ ) degrades Correctness to 0.6430 and Relevancy to 0.5815. Thus, structural graph grounding combined with governed intent routing provides effective anchors against relation hallucination and decision drift.

**Answering RQ3 (Representation-Aware Retrieval Effectiveness):** The configuration sequence reveals that neural reranking principally improves early-rank ordering rather than evidence coverage: E3-to-E4 gains are concentrated in Hit@1 and MRR@10, while the full stack does not surpass the lexical baseline on P@5 or Recall@5. Domain stratification further shows that E5’s additional Hit@1 benefit is concentrated in Financial queries; General queries improve over E1 by E4 and then remain unchanged, while neither Academic nor Scholarship improves over E1. This pattern is consistent with BM25 remaining competitive on exact course codes, cohort identifiers, and administrative terminology. Because E5 changes intent governance, graph grounding, and parent-child mapping together, these results compare complete configurations and do not attribute the observed gains causally to routing alone.

#### 4.5 Threats to Validity and Limitations

The administrative taxonomy and retrieval results are evaluated only at Can Tho University; multi-institutional validation remains planned. Domain-stratified results are descriptive and imbalanced (Academic  $N = 3$ , Financial  $N = 24$ , Scholarship  $N = 5$ , and General  $N = 18$ ), so the small Academic and Scholarship strata do not support claims of uniform cross-domain improvement. Controlled computational-efficiency evaluation and model compression remain future work. Furthermore, due to budgetary constraints, Gemini 2.5 Flash Lite was employed for both response generation and automated judging, which introduces potential model-specific evaluation bias. Curricula graphs rely on offline schemas, and automated judges penalize valid abstentions, underestimating practical correctness. Finally, while Scenario 3 validates bounded execution at the isolated decision level, fault injection across external infrastructure (e.g., Neo4j or Qdrant drops) represents orthogonal resilience evaluation reserved for future work.

### 5 Conclusion and Future Work

This paper addressed knowledge heterogeneity in university counseling by introducing CTU-CHAT, a centralized supervisor-routed multi-agent architecture coordinating four domain specialists with isolated evidence spaces and asymmetric tool privileges. By allocating relational curricula to Neo4j, statutory tuition to structured catalogs with deterministic arithmetic tools, and narrative decrees to hybrid RAG (BM25, dense embeddings, RRF, and neural reranking), CTU-CHAT eliminates the uniform representation bottleneck and decision-space pollution. Empirical evaluations on an official Can Tho University benchmark substantiate this design: attaining 97.0% supervisor routing accuracy with 100.0% bounded safety, superior retrieval performance (0.9600 Hit@1, 0.9700 MRR@10), and top factual exact match (62.03% Fact EM) in end-to-end generation. Future work will investigate cross-institutional adaptation, controlled computational-efficiency evaluation and model compression, and live longitudinal student trials.

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